Since there are four times as many samples, the number of bits per sample can be reduced from 16 bits to 14 bits, without degrading the sound quality. The $\sin(x)/x$ correction needed to compensate for the zeroth order hold of the DAC can be part of either the analog or digital filter.

Audio systems with more than one channel are said to be in **stereo** (from the Greek word for *solid*, or *three-dimensional*). Multiple channels send sound to the listener from different directions, providing a more accurate reproduction of the original music. Music played through a monaural (one channel) system often sounds artificial and bland. In comparison, a good stereo reproduction makes the listener feel as if the musicians are only a few feet away. Since the 1960s, high fidelity music has used two channels (left and right), while motion pictures have used four channels (left, right, center, and surround). In early stereo recordings (say, the Beatles or the Mamas And The Papas), individual singers can often be heard in only one channel or the other. This rapidly progressed into a more sophisticated **mix-down**, where the sound from many microphones in the recording studio is combined into the two channels. Mix-down is an art, aimed at providing the listener with the perception of *being there*.

The four channel sound used in motion pictures is called **Dolby Stereo**, with the home version called **Dolby Surround Pro Logic**. ("Dolby" and "Pro Logic" are trademarks of Dolby Laboratories Licensing Corp.). The four channels are encoded into the standard left and right channels, allowing regular two-channel stereo systems to reproduce the music. A Dolby decoder is used during playback to recreate the four channels of sound. The left and right channels, from speakers placed on each side of the movie or television screen, is similar to that of a regular two-channel stereo system. The speaker for the center channel is usually placed directly above or below the screen. Its purpose is to reproduce speech and other visually connected sounds, keeping them firmly centered on the screen, regardless of the seating position of the viewer/listener. The surround speakers are placed to the left and right of the listener, and may involve as many as twenty speakers in a large auditorium. The surround channel only contains midrange frequencies (say, 100 Hz to 7 kHz), and is *delayed* by 15 to 30 milliseconds. This delay makes the listener perceive that speech is coming from the screen, and not the sides. That is, the listener hears the speech coming from the front, followed by a delayed version of the speech coming from the sides. The listener's mind interprets the delayed signal as a reflection from the walls, and ignores it.

Companding

The data rate is important in telecommunication because it is directly proportional to the *cost* of transmitting the signal. Saving bits is the same as saving money. **Companding** is a common technique for reducing the data rate of audio signals by making the quantization levels *unequal*. As previously mentioned, the loudest sound that can be tolerated (120 dB SPL) is about one-million times the amplitude of the weakest sound that can be detected (0 dB